

9 (short)

- (a) Consider the halite structure AB shown in Fig. 7a. Find the ratio of the diameters of A and B ions such that the A and B ions are touching on the edges of the unit cell and the A ions are touching on the face diagonal.

[4]

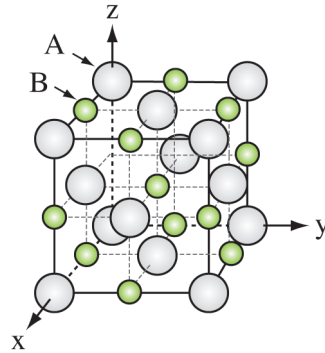


Fig. 7a

- (b) Figure 7b shows the unit cell of diamond cubic (DC) silicon carbide. The lattice constant of silicon carbide is $a = 0.436 \text{ nm}$. The atomic mass of silicon is $28.09 \text{ kg kmol}^{-1}$ and that of carbon is $12.01 \text{ kg kmol}^{-1}$. Calculate the theoretical density of the silicon carbide.

[6]

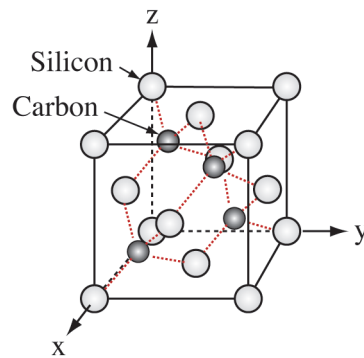


Fig. 7b